

Cybersecurity Bayesian Network: Phishing Detection

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1 Task 1 - Build a Bayesian Network in pyAgrum

1.1 Introduction

Phishing remains one of the most prevalent forms of cybercrime and continues to pose a significant threat to both individuals and organizations. Attackers typically disguise fraudulent emails as legitimate communications in an attempt to deceive users into revealing sensitive information, installing malware, or performing actions that compromise security. Despite advances in technical detection methods, phishing attacks remain highly successful because that exploit human behavior through social engineering techniques rather than purely technical vulnerabilities [AHNK21, JLN25].

Several studies have shown that phishing emails often contain identifiable indicators, such as suspicious sender information, misleading hyperlinks, poor, grammar, and language designed to create urgency or invoke authority [PBP⁺16, Nai15, Ros25]. At the same time, user-related factors such as cybersecurity awareness strongly influence whether a phishing attempt ultimately succeeds [ZBDR⁺24].

Because phishing detection involves uncertainty and incomplete information, Bayesian Networks provide a suitable modelling framework. They allow probabilistic reasoning based on observable evidence while explicitly representing causal relationships between variables. Rather than making a deterministic decision, a Bayesian Network can estimate the likelihood that an email is malicious and how different pieces of evidence contribute to that assessment.

The objective of this work is to construct a Bayesian Network capable of estimating the probability that an email is a phishing attempt and the probability that it ultimately leads to account compromise. The model combines observable characteristics of an email with behavioral characteristics of the user, enabling both prediction and explanation of phishing-related risks.

1.2 Methods

1.2.1 Variable Selection

The variables included in the network were selected based on commonly reported phishing indicators in cybersecurity literature. Research has identified sender legitimacy, hyperlinks, grammatical quality, and urgency cues as important characteristics that users rely on when distinguishing phishing emails from legitimate communications [PBP⁺16, Nai15, Ros25].

The Bayesian Network consist of eight binary variables, shown in Table 1. All variables were modelled as binary variables in order to reduce model complexity and facilitate interpretation of the resulting probability distributions.

Variable	Description	Values
SenderReputation	Indicates whether the sender appears trustworthy or suspicious.	good / suspicious
EmailGrammar	Represents the quality of grammar and spelling used in the email.	good / poor
ContainsLinks	Indicates whether the email contains hyperlinks.	yes / no
UrgencyLanguage	Indicates whether the email contains urgent or threatening language.	yes / no
PhishingEmail	Indicates whether the email is a phishing attempt.	yes / no
UserAwareness	Represents the cybersecurity awareness level of the user	high / low
ClickLink	Indicates whether the user clicks a link contained in the email.	yes / no
AccountCompromised	Indicates whether the user’s account becomes compromised.	yes / no

Table 1: Description and values of the selected variables in the Bayesian Network

1.2.2 Causal Structure

The network structure was manually designed according to causal assumptions derived from the phishing process.

SenderReputation, EmailGrammar, ContainsLinks, and UrgencyLanguage were modelled as direct causes of the variable PhishingEmail. This reflects the assumption that these observable characteristics provide evidence regarding whether an email is malicious. Previous studies have shown that phishing attacks frequently rely on suspicious sender identities, embedded links, and urgency-based persuasion strategies to increase user compliance [JLN25, Nai15, Ros25].

The variable ClickLink depends on both PhishingEmail and UserAwareness. This relationship captures the idea that phishing emails are more likely to trigger user interaction, while users with higher cybersecurity awareness are generally better able to recognize and avoid phishing attempts [ZBDR⁺24].

Finally, AccountCompromised was modelled as a direct consequence of ClickLink. This reflects a simplified attack chain in which account compromise becomes substantially more likely after interaction with a malicious link.

The resulting Bayesian Network models the progression from email characteristics to phishing detection, user behavior, and ultimately account compromise.

1.2.3 Network Structure Representation

The resulting Bayesian Network consists of eight nodes and seven directed edges. A graphical representation of the network is shown in Figure 1.

The network follows a simplified phishing attack chain. Email characteristics influence the probability that an email is a phishing attempt. The phishing status of the email, together with the cybersecurity awareness of the user, influences the probability that a link is clicked. Finally, clicking on a malicious link increases the probability of account compromise.

This structure was intentionally kept relatively compact in order to maintain interpretability while still capturing the most important causal relationships involved in phishing attacks.

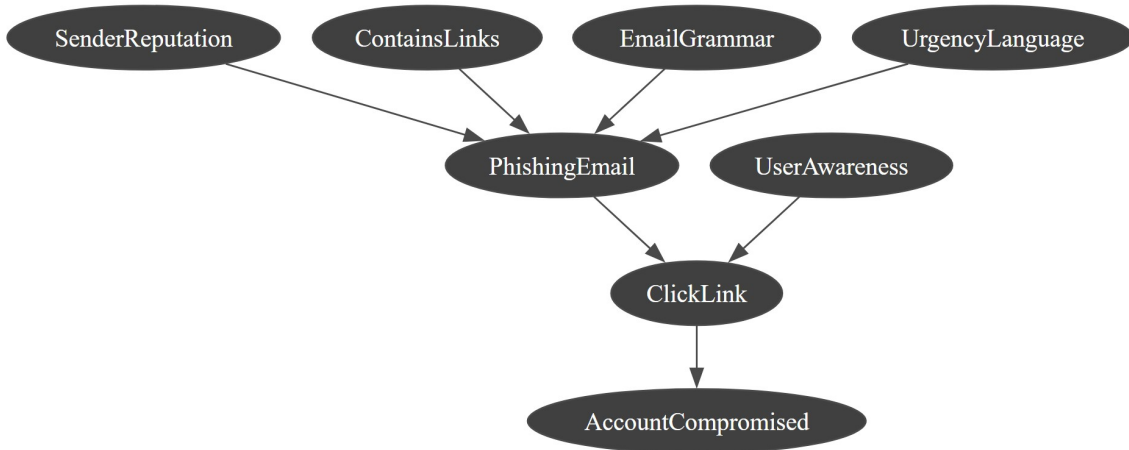


Figure 1: Bayesian Network structure for phishing detection and risk assessment

1.2.4 Parameter Specification

After defining the network structure, probability distributions were assigned to all variables. Since no complete dataset describing the selected domain was available, the parameters were specified manually using expert knowledge and observations reported in cybersecurity literature.

Prior probabilities were assigned to all root nodes. These probabilities reflect general assumptions about the prevalence of suspicious senders, poor grammar, embedded hyperlinks, urgency-based language, and cybersecurity awareness within a typical email environment.

For the variable *PhishingEmail*, a conditional probability table (CPT) was constructed based on the assumption that suspicious senders, poor grammar, embedded hyperlinks, and urgency cues all increase the likelihood of an email being a phishing attempt. Combinations containing multiple phishing indicators were therefore assigned substantially higher probabilities than combinations containing only legitimate characteristics.

The CPT of *ClickLink* was designed to reflect user susceptibility. Users with low cybersecurity awareness were assumed to be more likely to interact with links, particularly when confronted with phishing emails. Conversely, users with high awareness were assumed to be more resistant to such attacks.

Finally, the CPT of *AccountCompromised* reflects a simplified attack process in which account compromise becomes significantly more likely after a user clicks on a malicious link.

The complete CPTs used in the model are documented in the project repository¹ and were subsequently implemented in pyAgrum² for inference and data generation (see Task 2).

1.3 Implementation

The Bayesian Network was implemented using the Python library pyAgrum. The model structure was first defined by explicitly adding all variables as nodes, followed by the specification of directed edges representing causal relationships between variables.

After defining the network structure, conditional probability tables (CPTs) were manually assigned based on the expert-driven design described in the previous sections. These CPTs were encoded directly in pyAgrum using discrete probability distributions for each node.

The implemented model was then used for probabilistic inference using the LazyPropagation algorithm provided by pyAgrum. This allowed the computation of posterior probabilities under different evidence scenarios, which were used to validate the behavior of the network.

Finally, the model was exported to a `.bif` file format to ensure reproducibility and to enable its use in subsequent structure learning experiments in Task 2.

1.4 Validation and Analysis

The Bayesian Network was validated through a series of inference experiments designed to verify whether the resulting posterior probabilities were consistent with expected phishing behavior.

¹<https://github.com/kyanamarckx/bayesian-network/tree/main>

²<https://pyagrum.gitlab.io>

First, the prior probability of *PhishingEmail* was computed without introducing any evidence. The network estimated a baseline phishing probability of 19.31%. This value serves as a reference point for evaluating the influence of additional evidence.

To assess the impact of individual phishing indicators, the sender reputation was set to *suspicious*. Under this condition, the posterior probability of *PhishingEmail* increased from 19.31% to 50.92%. This substantial increase confirms that the network correctly interprets suspicious senders as an important indicator of phishing activity.

A second experiment combined all major phishing indicators considered in the model: suspicious sender reputation, poor grammar, embedded hyperlinks, and urgency-based language. When all four indicators were observed simultaneously, the posterior probability of *PhishingEmail* increased to 95%. This result demonstrates that the network successfully aggregates multiple sources of evidence and strongly favors the phishing hypothesis when numerous warning signs are present.

The effect of cybersecurity awareness was evaluated by comparing the probability that a user clicks a link contained in a phishing email. For users with high cybersecurity awareness, the probability of clicking a phishing link was 60%. In contrast, users with low cybersecurity awareness exhibited a click probability of 90%. These results reflect the intended modelling assumption that awareness reduces susceptibility to phishing attacks and confirm that the human factor is appropriately represented in the network.

Finally, an end-to-end attack scenario was analyzed by combining all phishing indicators with a user exhibiting low cybersecurity awareness. Under these conditions, the probability of account compromise reached 61.03%. This result illustrates how risk propagates through the causal chain represented by the Bayesian Network. Although the phishing probability of account compromise remains lower because successful compromise additionally depends on user interaction and the conditional probability of compromise after clicking a malicious link.

Overall, the inference results are consistent with the intended semantics of the model and no unexpected probabilistic behavior was observed. The experiments provide evidence that the Bayesian Network offers a plausible representation of phishing attacks and their potential consequences.

2 Task 2 - Learn a Bayesian Network in pyAgrum

2.1 Introduction

In this task, structure learning algorithms for Bayesian Networks are investigated using synthetic data generated from the original phishing detection model. The goal is to evaluate how well different learning approaches can recover the underlying network structure and how this affects classification performance.

In addition, the learned networks are compared with the original manually designed model and a Naive Bayes classifier in a predictive setting.

2.2 Methods

2.2.1 Data generation

Synthetic datasets were generated by sampling from the original Bayesian Network using pyAgrum. Three datasets sizes were used (100, 500, and 1000 samples) to study the effect of sample size on structure recovery. A separate training and test split (100 samples each) was used for classification experiments.

2.2.2 Structure learning methods

Two structure learning approaches were evaluated:

- **Search-and-score learning:** A greedy hill climbing algorithm using the Bayesian Information Criterion (BIC) as scoring function.
- **Constraint-based learning:** A dependency-based approach using mutual information (MIIC) to infer conditional dependencies between variables.

2.2.3 Evaluation of structure learning

The learned network structures were compared to the original Bayesian Network by analyzing the number of correctly recovered edges, missing edges, and additional incorrect edges.

2.2.4 Classification setup

Three models were evaluated in a classification setting:

- Original Bayesian Network (expert-defined structure)
- Learned Bayesian Network (best-performing model: constraint-based, 500 samples)
- Naive Bayes classifier (baseline model)

The class variable was *PhishingEmail*, and the remaining variables were used as features. Performance was evaluated using ROC curves and Area Under the Curve (AUC).

2.3 Results

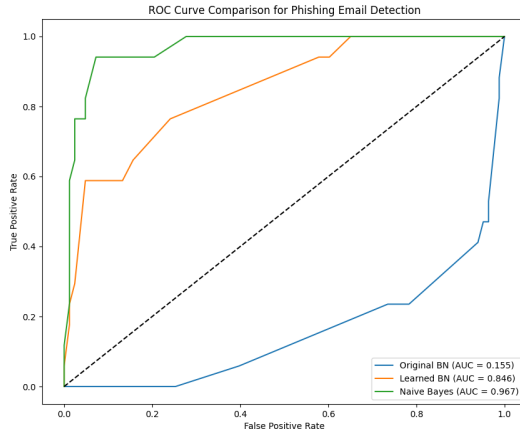
2.3.1 Structure learning results

The results show that structure recovery improves significantly with increasing sample size. The constraint-based approach consistently produced more accurate network structures than the search-and-score based method. With 500 samples (as well as the 1000 samples variant), the learned network closely resembled the original structure.

2.3.2 Classification results

The classification performance of the three models is summarized in Table 2 and visualized in Figure 2.

Interestingly, the Naive Bayes classifier achieved the highest performance despite its simplifying independence assumptions. The learned Bayesian Network significantly improved upon the original structure in terms of predictive performance, indicating that structure learning successfully captures useful dependencies from data.



Model	AUC
Original Bayesian Network	0.155
Learned Bayesian Network	0.846
Naive Bayes classifier	0.967

Table 2: AUC scores for the models

Figure 2: ROC Curves for the models

2.4 Conclusion

This experiment demonstrates that Bayesian Network structure learning is highly dependent on sample size and learning strategy. While the original expert-defined model provides interpretability, the learned models can significantly improve predictive performance. However, simpler models such as Naive Bayes may still outperform both in classification tasks when feature dependencies are strong and well-separated.

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